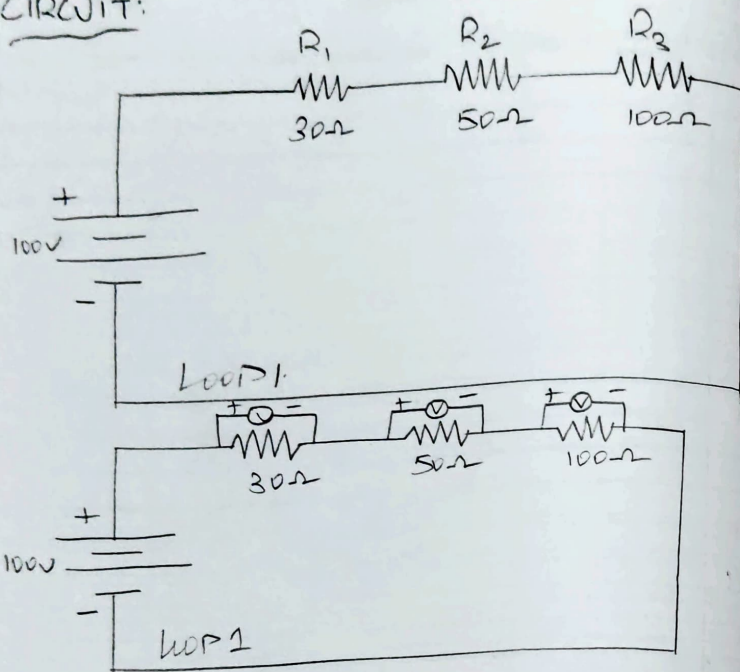


CIRCUIT:



TABULATION:

KVL	Source (V)	$V_{30\Omega}$ (V)	$V_{50\Omega}$ (V)	$V_{100\Omega}$ (V)	$V_{30\Omega} + V_{50\Omega} + V_{100\Omega}$ (V)
Theory	100	16.68V	27.8V	55.6V	100.05 V
Practical	100	16.7V	27.8V	55.6V	100.1 V

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VERIFICATION OF KIRCHOFF'S VOLTAGE LAW AND KIRCHOFF'S CURRENT LAW:

AIM:

To Verify Kirchhoff's voltage law for loop 1 and Kirchhoff's current law in node A for the following circuit.

APPARATUS REQUIRED:

Laptop with Proteus software.

THEORY:

According to Kirchhoff's voltage law sum of potential rises are equal to sum of potential drop in a loop. In loop 1 by KVL:

$$V = V_1 + V_2 + V_3 \quad \text{--- (1)}$$

$$100 = IR_1 + IR_2 + IR_3$$

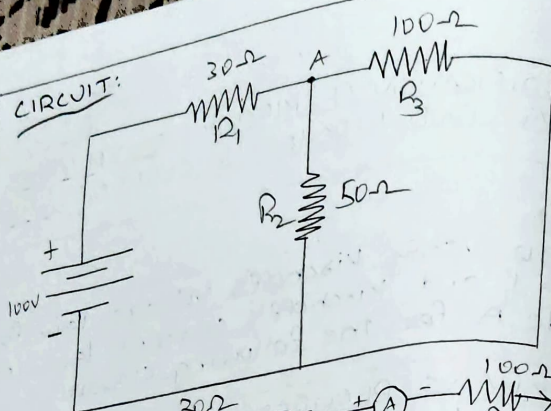
$$R_{eq} = R_1 + R_2 + R_3$$

$$R_{eq} = 30 + 50 + 100$$

$$R_{eq} = 180 \Omega$$

$$I = \frac{V}{R_{eq}} = \frac{100}{180} = \underline{\underline{0.556 \text{ A}}}$$

CIRCUIT:



TABULATION:

KLL	$I_{30\Omega}$ (A)	$I_{50\Omega}$ (A)	$I_{100\Omega}$ (A)	V_A (V)
Theoretical	1.48A	1.85A	0.56A	52.63 V
Practical	1.58A	1.05A	0.53A	52.6 V

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$$V_1 = V_{30\Omega} = IR_1 = 0.556 \times 30 = 16.68V$$

$$V_2 = V_{50\Omega} = IR_2 = 0.556 \times 50 = 27.8V$$

$$V_3 = V_{100\Omega} = IR_3 = 0.556 \times 100 = 55.6V$$

Put V_1, V_2, V_3 values in ①

$$100 = 16.68 + 27.8 + 55.6$$

$$100V = 100V$$

Hence Proved.

KIRCHOFF'S CURRENT LAW:

In a junction incoming current is equal to the outgoing current.

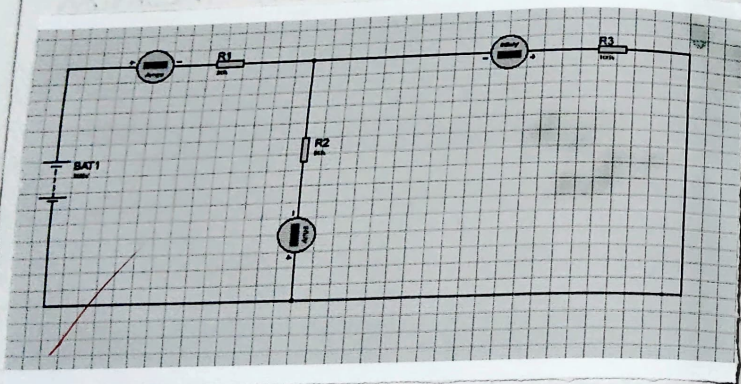
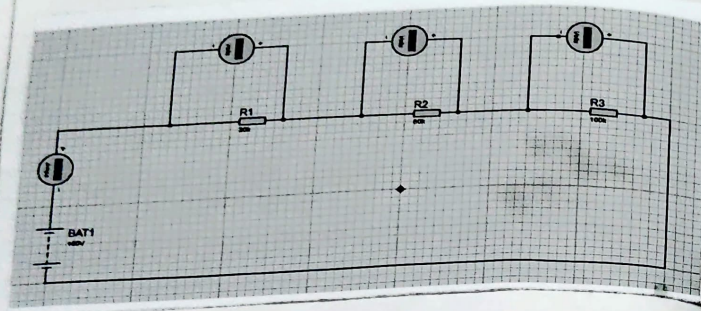
At node A:

$$I_{30\Omega} + I_{50\Omega} + I_{100\Omega} = 0$$

$$\frac{V-100}{30} + \frac{V}{50} + \frac{V}{100} = 0$$

$$V \left[\frac{1}{30} + \frac{1}{50} + \frac{1}{100} \right] - \frac{100}{30} = 0$$

$$V[0.0633] - 3.333 = 0$$



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$$V = \frac{3.333}{0.0633}$$

$$V = 52.63 \text{ V}$$

$$I_{30\Omega} = \frac{V - 100}{30} = \frac{52.63 - 100}{30} = -1.58 \text{ A}$$

$$I_{50\Omega} = \frac{V}{50} = \frac{52.63}{50} = 1.05 \text{ A}$$

$$I_{100\Omega} = \frac{V}{100} = \frac{52.63}{100} = 0.53 \text{ A}$$

At node A:

$$I_{30\Omega} + I_{50\Omega} + I_{100\Omega} = 0$$

$$-1.58 + 1.05 + 0.53 = 0$$

Hence Proved

RESULT:

Thus the Kirchhoff's voltage law and Kirchhoff's current law has been verified successfully.